Project 2: Numerical Solutions to Incompressible Navier–Stokes Equations

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Write a concise abstract here. State the governing equations solved, the numerical scheme implemented (fractional-step method with second-order Adams–Bashforth advection and Crank–Nicolson diffusion on a staggered grid), the Reynolds numbers investigated, and key findings regarding accuracy and physical behavior of the lid-driven cavity flow.

I. INTRODUCTION

The lid-driven cavity flow problem is a computational fluid dynamics (CFD) benchmark problem that consists of simulating the flow of fluid within a square domain with three rigid walls and a driving velocity on the top boundary. The top lid moves at a consant velocity, while the other three walls have a no-slip boundary condition. This setup creates a complex flow pattern with a primary vortex in the center and secondary vortices in the corners, which evolve as the Reynolds number increases. The problem is widely used to validate numerical methods for incompressible flows due to its well-documented solutions and sensitivity to numerical accuracy. The fractional stepping method used in this project allows for seperation of the velocity and pressure fields, which is crucial for solving the incompressible Navier–Stokes equations efficiently.

The objective of this project is to implement a fractional-step solver for the 2-D incompressible Navier–Stokes equations on a staggered grid, validate the results against the benchmark data provided by Ghia et al. for a range of Reynolds numbers.

II. GOVERNING EQUATIONS

The dimensionless 2-D incompressible Navier–Stokes equations are

$$\frac{\partial \boldsymbol{q}}{\partial t} + \boldsymbol{q} \cdot \nabla \boldsymbol{q} = -\nabla p + \frac{1}{Re} \nabla^2 \boldsymbol{q}, \tag{1}$$

$$\nabla \cdot \boldsymbol{q} = 0, \tag{2}$$

where $\boldsymbol{q} = \{u, v\}$ is the velocity field, p is pressure, and Re is the Reynolds number. Describe the boundary conditions (no-slip on three walls; lid velocity $u_{\text{top}} = 1$ on the top boundary).

III. OPERATORS

III.A. Gradient Operator

The gradient operator ($\mathbf{G}$) is discretized using central differences for the interior points and one-sided differences for the boundary points. It maps pressure difference to velocity face values using the following formulas:

$$(\mathbf{G}p)^u_{i,j} = \frac{p_{i,j} - p_{i-1,j}}{\Delta x}, \quad (\mathbf{G}p)^v_{i,j} = \frac{p_{i,j} - p_{i,j-1}}{\Delta y} \tag{3}$$

In this case, the gradient operator is second order accurate in the interior.

III.B. Divergence Operators

The divergence operator ($\mathbf{D}$) is discretized using central differences for the interior points and one-sided differences for the boundary points. It maps velocity face values to cell-centered divergence. The central points are calculated using the following formula:

$$(\mathbf{Du})_{i,j} = \frac{u_{i+1,j} - u_{i,j}}{\Delta x} + \frac{v_{i,j+1} - v_{i,j}}{\Delta y} \tag{4}$$

The full divergence operator is split into the sum of the interior and boundary contributions. The boundary contribution ($\mathbf{bc}_D$) at each wall is given in the following table:

III.C. Laplacian Operators

For the interior points:

$$(\mathbf{Lu})_{i,j} = \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta y^2} \tag{5}$$

III.D. Advection Operator

III.E. Unit Testing of Operators

Table 1 summarizes the unit tests performed for each discrete operator, verifying their correctness and expected convergence properties. All tests passed successfully, confirming the accuracy of the implemented operators. With these results in mind, we can now move on to using the solver on the lid-driven cavity flow problem.

IV. NUMERICAL METHOD

IV.A. Staggered Grid Formulation

The staggered grid arrangement is illustrated in Figure 1, where the horizontal velocity u is defined at the vertical cell faces, the vertical velocity v is defined at the

TABLE I Unit tests for discrete operators, verifying correctness of each component.

Test	Operator	What is checked	Expected result	Pass?
G1	$\mathbf{G}$	Gradient of linear p	Exact (machine ε)	✓
G2	$\mathbf{G}$	Gradient of quadratic p	Error $O(\Delta x^2)$	✓
D1	$\mathbf{DG}$	Composition $\approx \nabla^2$	Error $O(\Delta x^2)$ interior	✓
D2	$\mathbf{D}$	Divergence of div-free field	Error $O(\Delta x^2)$ interior	✓
D3	$\mathbf{D}$, $\mathbf{G}$	Adjoint: $\boldsymbol{u}^\top \mathbf{G}\boldsymbol{p} = \boldsymbol{p}^\top \mathbf{D}\boldsymbol{u}$	Machine ε	✓
BC1	$\mathbf{bc}_D$	Global mass balance $\sum \mathbf{bc}_D = 0$	Exact	✓
BC2	$\mathbf{bc}_L$	Top-wall BC value $= 2U^*/\Delta y^2$	Exact	✓
L1	$\mathbf{L}$	Laplacian of quadratic field $= 2$	Exact interior	✓
L2	$\mathbf{L}$	Laplacian of linear field $= 0$	Machine ε	✓
L3	$\mathbf{L}$	Self-adjoint: $\boldsymbol{u}^\top \mathbf{L}\boldsymbol{v} = \boldsymbol{v}^\top \mathbf{L}\boldsymbol{u}$	Machine ε	✓
A1	$\mathbf{A}$	Uniform flow gives zero advection	Machine ε	✓
A2	$\mathbf{A}$	Energy: $\boldsymbol{u}^\top \mathbf{A}(\boldsymbol{u}) \approx 0$	Small ($O(\Delta x^2)$)	✓
CG1	CG	Solves Poisson problem to tolerance	$<$ tol	✓
FULL	$\mathbf{T}$	$\boldsymbol{p}^\top \mathbf{T}\boldsymbol{p} > 0$ (SPD)	Positive	✓

horizontal cell faces, and the pressure p is defined at the cell centers. This helps to avoid incorrect pressure calculations and ensures a consistent discretization of the divergence and gradient operators. The pressure at the bottom-left cell is pinned to zero to provide a reference for the pressure field.

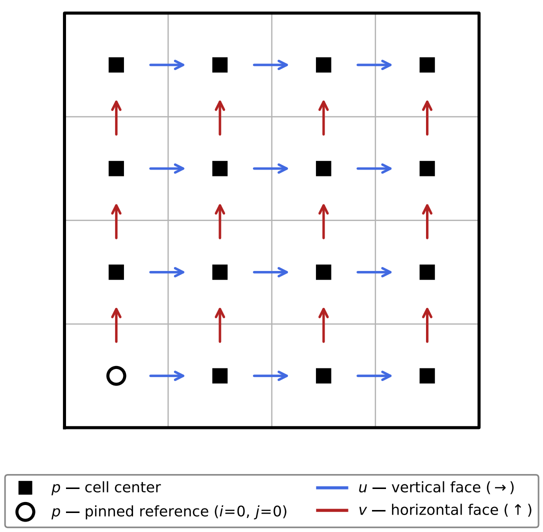

FIG. 1 Staggered grid formulation. Horizontal velocity u ($\longrightarrow$), vertical velocity v ($\uparrow$), and pressure p (■) are located at cell faces and cell centers, respectively. The open circle (○) marks the pinned pressure reference.

IV.B. Spatial Discretization

Introduce the discrete divergence (D), gradient (G), advection (N), and Laplacian (L) operators. State their orders of accuracy and discretization stencils.

IV.C. Time Integration: Fractional-Step Method

Using second-order Adams–Bashforth for advection and Crank–Nicolson for diffusion, the semi-discrete momentum equation and continuity constraint form the saddle-point system[?]

$$\begin{pmatrix} A & G \\ D & 0 \end{pmatrix} \begin{pmatrix} \boldsymbol{q}^{n+1} \\ p^{n+1} \end{pmatrix} = \begin{pmatrix} \boldsymbol{r}^n + \mathrm{bc}_1 \\ 0 \end{pmatrix}, \tag{6}$$

where $A = I/\Delta t - L/(2Re)$ and $\boldsymbol{r}^n$ contains the explicit Adams–Bashforth advection and Crank–Nicolson diffusion terms. Applying block LU decomposition and approximating $A^{-1} \approx \Delta t\, I$ yields the three-step fractional-step algorithm:

$$A\boldsymbol{q}^* = \boldsymbol{r}^n + \mathrm{bc}_1, \tag{7}$$

$$D\,\Delta t\, G\, p^{n+1} = D\boldsymbol{q}^*, \tag{8}$$

$$\boldsymbol{q}^{n+1} = \boldsymbol{q}^* - \Delta t\, G\, p^{n+1}. \tag{9}$$

The intermediate velocity system (7) and pressure Poisson equation (8) are solved at each time step using a conjugate gradient method[?] .

FIG. 2 (a) Temporal convergence versus CFL and (b) spatial convergence versus grid size N for v at the cavity center. Reference lines indicate the expected convergence rates.

IV.D. Time Step and Stopping Criterion

The time step is chosen as

$$\Delta t = \min\left(\mathrm{CFL}\, \frac{\Delta x}{u_{\mathrm{top}}},\ \mathrm{CFL}\, (\Delta x)^2 Re \right), \tag{10}$$

and time integration proceeds until the steady-state criterion $\max |\boldsymbol{q}^{n+1} - \boldsymbol{q}^n| < 10^{-8}$ is satisfied.

V. RESULTS

V.A. Grid and Temporal Convergence

Present log–log convergence plots for vertical velocity v at the cavity center $(x, y) = (0.5, 0.5)$.

V.B. Validation Against Ghia et al.

Figure 3 compares centerline velocities and top-wall vorticity to the reference data of Ghia, Ghia, and Shin[?] for $Re = \{100, 400, 1000, 3200, 5000\}$.

FIG. 3 (a) Horizontal velocity u along the vertical centerline $x = 0.5$, (b) vertical velocity v along the horizontal centerline $y = 0.5$, and (c) vorticity ω along the top wall $y = 1.0$ for several Reynolds numbers. Solid lines: present simulation; symbols: Ghia et al.[?] .

FIG. 4 Streamline contours at $Re = 100$ (top), 1000 (middle), and 3200 (bottom). Left column: present simulation; right column: Ghia et al.[?] .

V.C. Streamline Topology

Figure 4 shows streamline patterns for $Re = \{100, 1000, 3200\}$. Discuss the evolution of the primary vortex center and the growth of secondary corner vortices with increasing Reynolds number.

V.D. Vorticity Contours

Present vorticity contours and compare qualitatively and quantitatively to the reference[?] . Discuss boundary-layer thinning at higher Re and any limitations of the uniform-grid approach.

V.E. Additional Exploration

Present additional results that go beyond the required validation. For example, ...

VI. DISCUSSION

Discuss the observed convergence rates in light of the numerical scheme, the accuracy of results at varying Reynolds numbers, and physical observations from the flow field. Address any discrepancies with the reference data.

VII. CONCLUSION

Summarize the key findings of the project in 4–6 sentences. State clearly the validated Reynolds number range, convergence rates observed, and any limitations identified.

NOTE ON CODE SUBMISSION

Your MATLAB/Python code should be appended *after* the references as a separate PDF printout. It does not count toward the 10-page limit and does not need to appear in the LaTeX source.